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SUPERSEDING
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MILITARY SPECIFICATION

FURNACES, HEAT-TREATING, AND QUENCHING, IN AND OUT, AND STRAIGHT THROUGH DESIGN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers both semi-automatic and automatic controlled atmosphere, heat-treating furnaces of the horizontal box, in and out, and straight through design, with integral oil quench and atmosphere cooling system.

1.2 Classification. The furnaces covered by this specification shall be of the following types, styles, and classes as specified (see 6.2.1).

Type I - In and Out Design

Type II - Straight Through Design
  Style A - Semi- Automatic
  Style B - Automatic
    Class 1 - Gas Fired Radiant Tube
    Class 2 - Electrically Heated

1.2.1 Sizes. The size of the furnace shall be specified (see 3.6 and 6.2.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3424

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2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-3296	Forges, Furnaces, and Ovens (Exclusive of Space Heating and Cooking), Packaging of
MIL-H-6875	Heat Treatment of Steels (Aerospace Practice, Process for)

STANDARDS

FEDERAL

* FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

Title 29, Code of Federal Regulations, Chapter, XVII, Part 1910, and Amendments.	Occupational Safety and Health Administration Standards

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

* AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

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C1 - National Electrical Code
C96.1 - Temperature Measurement Thermocouples

(Applications for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

FACTORY MUTUAL ENGINEERING CORPORATION (FM)

Handbook of Industrial Loss Prevention

(Application for copies should be addressed to Factory Mutual Engineering Corporation, 1151 Boston-Providence Turnpike, Norwood, Mass. 02062.)

3. REQUIREMENTS

* 3.1 First article. The supplier shall furnish one complete furnace of each type, style, class and size specified in the contract or order for first article inspection in accordance with Section 4 (see 6.4).

3.2 Design. Furnaces and equipment provided in accordance with this specification shall be new and of the manufacturer's current design. The furnaces shall be of the horizontal box design with integral oil quench and atmosphere cooling system. The furnace shall consist of two chambers, the heating chamber and cooling chamber. The cooling chamber shall contain the oil quench tank, elevator system, and atmosphere cooling compartment. The chambers shall be separated by a refractory insulated air or air-hydraulic operated door. The furnace shall be of gas-tight construction and designed for operation with an endothermic type of manufactured atmosphere. The furnace shall provide atmosphere protection for the workload through the complete work cycle. Unless otherwise specified, the operating temperature range of the furnace shall be from 1400 degrees F through not less than 1750 degrees F. When specified (see 6.2.1), the furnace shall operate through a temperature range of 1400 degrees F through 2000 degrees F.

3.2.1 Type I. The Type I furnace shall be of the "in and out" design with a single station located at the front of the furnace for loading and unloading. The cooling chamber shall be directly attached to the front of the heating chamber. The oil quench tank shall be located in the lower part of the cooling chamber and atmosphere cooling compartment located in the upper part of the cooling chamber, with a vestibule separating the quench tank and the atmosphere cooling compartment.

3.2.2 Type II. The Type II furnaces shall be of the "straight" through design with a loading platform at the front and an unloading platform at the rear. The cooling chamber shall be located at the rear of the furnace and shall be directly attached to the heating chamber.

3.2.3 Style A. The Style A furnace shall be semi-automatic and shall provide for manually transferring the charge from the loading table into the vesti-

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bule of the Type I furnace or into the heating chamber of the Type II furnace. After selecting the desired purge (Type I only), heat and quench cycles and sequence of operation, the furnace shall automatically process the charge through the preselected cycle sequence and provided an audiable signal when the charge is ready for unloading. When specified (see 6.2.1), manually actuated powered loading and unloading tables shall be provided.

* 3.2.4 Style B. The Style B furnace shall be fully automatic and shall provide for automatically transferring individual loads from the loading platform through the preselected purge, heat, optional quench and drain cycles and onto the unloading platform. The Type I, Style B furnace system shall provide for processing a second charge through the purge cycle while a charge is being heated. The second charge shall be transferred into the heating chamber, while a charge is being quenched, without breaking the atmosphere seal or opening the external door of the quench chamber. The Type II, Style B furnace shall automatically transfer a new charge into the heating chamber, after the heated charged has entered the cooling chamber. When specified (see 6.2.1), the Type II furnace shall have a front loading purge vestibule with automatic integral loading device and shall automatically transfer the charge into the heating chamber, after the heated charge enters the cooling chamber.

3.2.5 Class 1. The Class 1 furnace shall be heated by gas fired radiant tubes. Unless otherwise specified, Class 1 furnaces shall be designed to operate on a fuel supply of 1000 BTU/cu foot of natural gas. When required the gas fired furnace shall be designed and equipped to operate on either a butane or propane fuel supply as specified (see 6.2.1).

3.2.6 Class 2. The Class 2 furnace shall be electrically heated and, unless otherwise specified (see 6.2.1), designed for operation on a primary power supply of 460 volts, 60 Hz, 3 phase current.

* 3.2.7 Safety and health requirements. Each furnace shall be provided with safety features and devices for the protection of personnel, equipment and the work being processed. In addition to other safety requirements specified herein, the safety system of the furnace shall conform to the applicable requirements of FM "Handbook of Industrial Loss Prevention". The furnace shall comply with the standards promulgated under Occupational Safety and Health Act (OHSA) applicable to the furnace itself. Additional requirements for safety and health shall be as specified (see 6.2.1 and 6.5).

3.2.8 Lubrication. When necessary to reduce wear and maintain efficiency, means shall be provided for lubrication of all sliding and rotating parts, except non-additive pack-seal bearings. Lubrication systems shall be designed and built to prevent any contamination of atmosphere, components, material or equipment. Lubrication points shall be readily accessible and identified in the operators manual with the type of lubricant required and instructions for application.

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3.2.9 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts of furnace.

3.2.10 Accessibility. All parts subject to wear, breakage, or distortion and all parts which require periodic lubrication or maintenance shall be accessible for adjustments, replacement, and lubrication as applicable.

3.2.11 Material. All material used in the fabrication of the furnace and related equipment shall be of the quality necessary to produce a furnace to meet the requirements described herein. Materials that are exposed to the internal environment of the furnace chambers shall be compatible with the type of atmosphere and temperature conditions specified for each furnace.

* 3.2.12 Reclaimed materials. The furnaces may contain reclaimed materials provided such materials will not jeopardize the furnace intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical composition and physical properties as the materials originally selected for use on the furnace.

3.3 Construction. The furnace shall be constructed so that when installed and connected to the required supply sources, it can be used for any purpose described herein. All components and accessory equipment shall be new, and functionally suitable for the purpose intended. Repaired parts shall not be used.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. In no event shall such processes as welding, peening, plugging, or filling with solders or pastes be used on castings or forgings to reclaim any defective parts for use on the furnace.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where those operations are required in the original design. These operations shall not be employed as repair measures for defective parts.

3.3.3 Fastening devices. All screws, pins, bolts, and similar fasteners shall be installed with means for preventing loss of tightness. Fasteners subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled unless sharpness is required to perform a necessary function.

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3.3.5 Painting. The furnace shall be painted with heat resisting paint in accordance with the manufacturer's standard commercial practice.

3.3.6 Threads. All threaded parts shall conform to FED-STD-H28.

* 3.3.7 Dials. All dials shall be graduated. Dial diameters shall be such that the graduations may be easily read from the normal operating position. Dials shall be permanently and legibly engraved, etched, or marked on a non-glare background.

* 3.3.8 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embosed or stamped in boldface on a contrasting background.

* 3.4 Electrical equipment. The furnace shall be furnished complete with all necessary electrical motors, contactors, controls, relays, transformer, signal lights and switches pre-wired where possible for operation upon connection to source of power. All power actuated, manually adjusted, and automatic controls shall operate on a 115 volt, 60 Hz, power supply. All electrical equipment and wiring shall be in accordance with ANSI-C1.

* 3.4.1 Motors. Motors rated in excess of 1/2 horsepower shall be dual voltage, 230/460 volt, 60 Hz, 3 phase type, and unless otherwise specified (see 6.2.1), initially wired to operate on 460 volts. Starters and wiring shall be designed and sized to permit operation on 230 volts. Motors rated at 1/4 horsepower or less shall operate on a 115 volt, 60 Hz power supply.

3.4.2 Electrical connections. Electrical connections of conductors and terminal parts shall be either the pressure or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Flux for soldering shall be of the resin base type.

3.4.3 Electromagnetic interference control. When specified (see 6.2.1) the furnace shall have electromagnetic interference control in accordance with the applicable requirements of MIL-STD-461.

3.5 Performance. Each furnace shall be capable of continuous operation at maximum operating temperature with a charge equivalent to not less than 125 pounds of steel per square foot of hearth area. With the furnace charged and the temperature stabilized, the temperature variation from control point to any other point within the chamber working area shall not vary more than $\pm 15^{O}$F. Each furnace shall be capable of maintaining the temperature uniformity specified with any size charge up to maximum load and any desired temperature control point within the operating temperature range of the furnace. The

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TABLE I. Sizes, Performance Characteristics and Capacities.

Size, Furnace Working Dimensions In Inches			Recovery Rate Net Lbs. of Steel/Hr to 1550°F	Maximum Charge Net at 1750°F Not Less Than	Class 1 Gas Input 1000s BTU/HR	Class 2 KW Input	Oil Quench Tank Capacity Gallons Within ±5%
Width	Length	Height					
18	24	12	300	375	360	36	400
24	36	18	600	800	500	50	800
24	36	24	600	800	550	55	800
24	48	18	800	1000	800	70	1000
30	48	24	1000	1250	900	90	1250
30	48	30	1000	1800	1200	100	1700
36	48	24	1675	2000	1200	150	1200
36	48	30	1900	2500	1200	178	1500
36	48	36	2000	3000	1200	180	1800
36	72	24	2000	3000	1400	---	3000

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performance characteristics and capacity requirements specified in Table I for each size furnace shall be minimum requirements. If a size different from the sizes listed herein is required, the recovery rate shall be based on 100 lbs of steel/sq ft of hearth/hour to 1550°F, and a maximum load of 150 lbs/sq ft of hearth area.

3.6 Sizes. The size of the furnace work chamber, width, length and height shall be specified and should be selected from the dimensions listed in Table I. The dimensions specified shall be the minimum inside effective working area of the heating chamber in which performance requirements and temperature uniformity specified herein shall be maintained.

3.7 Components and systems.

3.7.1 Shell. The outer shell of both the heating chamber and the cooling chamber shall be fabricated of steel plate not less than 3/16 inch in thickness. The entire unit shall be reinforced with structural steel members and electrically welded to form an atmosphere tight unit. The working height of the furnace shall be between 44 and 50 inches above floor level.

3.7.2 Insulation. The lining of the furnace heating chamber shall consist of either at least 4-1/2 inches of 2300°F insulating refractory brick, backed up with not less than 4-1/2 inches of insulating block or preformed insulation, or 10 inches of 2300°F insulating fire brick. Powdered, shredded, or granular material shall not be used for insulating purposes. When mortar is used, it should be compatible with the material being cemented together.

3.7.3 Hearth. Design of the hearth shall permit positive atmosphere circulation throughout the work area. The hearth shall be capable of supporting not less than 200 pounds per square foot of hearth area at maximum temperature without permanent distortion. Design and construction of the hearth shall permit maximum loads to slide in and out, or through the heating chamber by means of a powered transfer mechanism without interference.

3.7.4 Doors. Doors shall be of the vertical lift, air or air-hydraulic type. Doors shall be fabricated or cast from a material which will resist distortion under maximum furnace operating cycles. Heating chamber doors shall be lined with 6 inches of fire brick and insulating block not less than 4-1/2 inches of which shall be fire brick. External doors shall provide a gas tight seal in the closed position.

3.7.4.1 Flame curtain. The Type I furnace shall be provided with a gas-air-flame curtain located below the vestibule door opening. The Type II furnace shall be provided with two gas-air-flame curtains, with one located below the heating chamber entrance door and one located below the cooling chamber discharge door. Each curtain shall operate automatically by limit

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switches through solenoid valves and a constant burning pilot to protect the atmosphere when the door is opened. An interlock system shall prevent the door from being opened, unless the pilot is proven.

3.7.5 Gas fired heating system (Class 1). Class 1 furnaces shall be heated by gas fired radiant tubes. The tubes shall completely separate the firing system from the furnace protective atmoshpere. The tubes shall not sag or deform when operated continuously at maximum furnace temperature. Each furnace shall be provided with all necessary burners, gas-air mixing systems, governors, manifolds, valves, blowers, gauges, pilots, safety devices and systems required for safe and efficient operation when connected to the specified supply source.

* 3.7.6 Heating elements (Class 2). Class 2 furnaces shall be heated by electric heating elements. The elements shall be designed and located in the heating chamber to meet the heating requirements and temperature uniformity specified for the furnace. The heating elements shall be either graphite type elements enclosed in sealed radiant tubes containing a protective nitrogen atmosphere or 70 percent nickel, 30 percent chromium alloy enclosed in radiant tubes and protected forom the chamber atmoshpere or nickel chromium alloy sheet designed to operate on a reduced voltage that will be compatible with the atmosphere composition specified for the furnace. The power density on the alloy type elements shall not exceed 15 watts per square inch of radiation surface area.

* 3.7.7 Oil quench system. Each quench tank shall be provided with a heat exchanger cooling system capable of limiting the quench oil temperature increase to a maximum of 30°F when quenching a maximum furnace load of steel at 1750°F. The cooling system shall be automatically controlled and capable of lowering the temperature of a capacity tank of quenching oil not less than 30°F in a time limit of not more than 2 hours. Unless otherwise specified, the cooling system shall be of the water cooled type with the heat exchanger and pump located external to the quench tank. When required (see 6.2.1), the heat exchanger shall be of the air cool type. The quench tank shall be provided with an electrical heating system. The heating system shall be capable of raising the temperature of the quenching oil from 100°F to 300°F in not more than 6 hours. The heating system shall be automatically controlled and shall maintain a temperature uniformity within the quench tank of plus or minus 25°F from any control setting between 75°F and 400°F. Each heated tank shall be insulated on all exposed sides. The outside surface temperature of the quench tank shall be compatible with the furnace outside shell temperature. The oil quench system shall assure conformance with the severity of quench requirements specified in MIL-H-6875. The quench tank shall be provided with a temperature indicating gage and an adjustable excess temperature audible warning system having a range of from 100°F to not less than 400°F.

3.7.8 Atmosphere cooling compartment. The atmosphere cooling compartment shall be cooled on not less than 2 sides. The compartment may be

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cooled by circulating oil discharged from the heat exchanger through cooling jackets located on the inside compartment walls. If water cooling is used, all water jackets or plate coils shall be located external to the atmosphere cooling compartment walls. The compartment shall be capable of accepting maximum loads discarded from the heating chamber and provide atmosphere cooling through the complete cooling cycle. The cooling compartment shall be provided with a fan to circulate the atmosphere through the work area.

3.7.9 Elevator. Each furnace quenching system shall be provided with an air or air-hydraulic operated elevator system. The elevator for Type I furnaces shall be double decked. Each deck shall be provided with rollers or skid rails. The elevator shall be capable of operating with a load equal to not less than 200 pounds/square foot of heating chamber area on an air pressure not greater than 100 PSIG.

3.7.10 Recirculating fan. Each furnace heating chamber shall be provided with one or more recirculating alloy steel type fan. The fan(s) shall be capable of circulating a volume of atmosphere equivalent to 25 cubic feet/minute/square foot of the heating chamber walls. Fan bearings mounted to the chamber housing shall be water cooled.

3.7.11 Load transfer mechanism. Load transfer systems shall be interlocked with the furnace doors to prevent their operation when the door is closed. Operation of the transfer system for the Type II (straight through) furnace shall provide for processing individual loads through the cycle with only one heating chamber door open at any one time.

* 3.7.12 Thermocouples. Thermocouples shall be used as the temperature sensing device for each furnace. Thermocouples shall conform to ANSI C96.1. One thermocouple shall be used with each indicating, recording, and controlling instrument and one with the excess temperature control instrument. The thermocouple lead wires shall be not less than 50 feet in length and of the same material as the thermocouples or of a material having the same thermo-electric characteristics.

3.7.13 Program control system. Each furnace shall be provided with a program control system integrated with the furnace instrument controls. The system shall provide for preselecting any desired soak time from 0-10 hours and quench time from 0-55 minutes with an option for oil quench or atmosphere cool. The soak time shall not begin until chamber temperature has recovered to within 15°F of control point. Furnaces having a purge vestibule shall provide for an optional 0-55 minutes purge cycle. Each furnace shall provide for an oil drain cycle of 0-55 minutes, following oil quench and before the charge is transferred to the unloading station. If additional

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or different cycles are required, requirements shall be as specified (see 6.2.1). Having once selected the desired cycle sequence, duplication shall be repeated continually. In addition to the automatic control system, each furnace shall be provided with manual over-ride controls located at the load station for use in testing or emergency conditions.

3.7.13.1 Instrument panel. Each furnace shall be provided with dust proof floor mounted instrument panels. The panels shall be for mounting the indicating and recording temperature controls, excess temperature control, contactors, switches, cycle timing devices and any other accessories necessary for efficiency and safe operation. The power contactor should be mounted in a panel separated from the instruments. Each furnace shall be provided with a series of signal lights to indicate the location of the charge in the furnace. Instrument panels shall provide for either semi-flush or flush mounting of all instruments.

3.7.13.2 Instruments. The furnace shall be provided with an indicating-recording-controller type instrument system conforming to the requirements of 3.6.13.3 through 3.6.13.4. Accuracy of the temperature control system shall be such that the chamber temperature at equilibrium, the maximum cumulative error of all elements in the system shall not exceed $\pm 25^{\circ}F$ between set control point and all points in the work chamber. Each indicating and recording instrument shall be marked with the particular type of thermocouple for which it is calibrated. Automatic reference junction compensation shall be provided in all instruments using thermocouples as the primary temperature sensing element.

* 3.7.13.3 Temperature recorder. The temperature recorder shall be of the strip chart, self-balancing type. The calibrated chart width shall be not be less than 9 inches, and the length shall be not less than 80 feet to the roll with a chart speed of one (1) inch per hour. The pen mechanism shall drive the pen across the entire calibrated chart width in not more than 30 seconds. A platen located immediately behind the chart plane shall be furnished to permit the marking of notations on the chart while the instrument is in operation. The chart mechanism shall be enclosed in a dust proof housing with a hinged door. The door shall be provided with a glass that will permit full view of the chart width and not less than 9 inches of chart length.

3.7.13.4 Temperature controller. The temperature controller shall be of the proportioning, automatic controlling type. The control indicator shall have a calibrated scale length of not less than 10 inches and a

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temperature range of 0 degrees F - 2000°F. The increments between successive temperature marks shall be not greater than 20°F with the spacing of numbered marks not greater than 400°F. The control indicator shall be provided with two pointers. One pointer shall be adjustable to the desired control temperature, and one pointer shall indicate the acutal furnace temperature.

* 3.7.13.5 Excess temperature control. The furnace shall be provided with an excess temperature controller. The excess temperature control circuit shall be independent of the control circuitry. The controller shall provide for selecting the over-temperature control point and shall function to shut off the heat source to the furnace when the over-temperature control point is reached. The control shall provide for furnace shut down in the event of thermocouple failure or failure of other components in the circuit. In case the furance is shut down by the excess temperature control system, it shall remain off until started manually. Calibrated accuracy shall be within one percent of full scale reading.

3.7.13.6 Thermocouple burn-out alarm signal. The excess temperature cutout and the temperature controller shall be designed so that loss of either thermocouple circuits shall provide an audible and visual alarm signal.

3.8 Atmosphere gas metering panel. When specified (see 6.2.1), the furnace shall be provided with an atmosphere gas metering panel. The panel shall include flow meters with flow indicators for measuring the flow of endothermic gas and additive gases such as ammonia and natural gas, including electric solenoid, manual and check valves as required to control gas flow.

3.9 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate shall be securely attached to each furnace. The nameplate shall contain the information listed below. If the furnace is a special model, the model designation shall include the model of the basic standard furnace and a suffix keyed to the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input characteristics and ratings
Fuel gas characteristics (Class 1 only)
Quench tank capacity (gallons)

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Inside heating chamber dimensions
Operating temperature range (degrees F)
Compatible atmosphere
Date of manufacture
Contract or order number
National Stock Number or Plant Equipment Code
U S

* 3.10 Technical data. Technical data shall be furnished as specified (see 6.3).

3.11 Workmanship. Workmanship of the furnace and accessories shall be commensurate with the requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be performed on the first article submitted in accordance with 3.1. First article inspection shall consist of all tests in 4.6.4 through 4.6.6. Failure of the first article to pass the examination or one or more first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on each item prior to being offered for acceptance under the contract. Quality conformance inspection shall consist of the examination in 4.5, the tests in 4.6.1 through 4.6.3 and the inspection in 4.7. Failure of the furnace to pass the examination, the tests, or the inspection shall be cause for rejection.

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4.5 Examination. The furnace and equipment shall be examined for design, safety compliance, dimensions, construction, material, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification.

* 4.6 Tests. All tests shall be conducted at an ambient temperature of 77 degrees F ±18 degrees F. Test instruments and sensing elements used to perform the tests specified for the furnaces and instruments shall have been checked against a standard potentiometer type instrument and calibrated thermocouple of known accuracy within the previous three months to read accurate within ± 2 degrees F.

4.6.1 Instrument test. Indicating and recording instruments shall be tested by impressing various predetermined voltages, simulating thermocouple outputs for different furnace temperatures to terminals where normally the thermocouples would be connected. These tests shall be made in accordance with, and results checked against, standards established in ANSI 96.1. The instruments shall be tested through their operating range. Each circuit shall be tested to determine compliance with 3.7.13.

4.6.2 Circuit test. Each circuit of the electrical system shall be tested for dielectric strength, continuity, insulation-resistance and faulty grounds in accordance with the requirements of ANSI-C1.

4.6.3 Operational test. Each furnace shall be operated "unheated" through not less than three different preselected cycles and all components and systems shall be adjusted as necessary to assure correct operation. All system capacity and dimensions shall be measured and checked for compliance with requirements specified for the size furnace under test. Each safety device or system shall be tested by simulating the failure that would normallly cause the safety device to function. Each system shall perform its intended function.

4.6.4 Performance test. The furnace under test shall be prepared for operation and operated in accordance with the manufacturer's handbook of instructions until equilibrium has been established and the refractory and insulation has cured, the controls shall be programmed to perform the following cycle sequence.

a. Purge, 20 minutes (Type I only) and Type II when equipped with a purge chamber.
b. soak period, 1 hour.
c. Oil quench, 40 minutes.
d. Oil drain, 15 minutes.

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With the chamber temperature stabilized at 1550°F, a charge of iron or steel equivalent to 100 pounds per square foot of heating chamber area shall be placed in the normal loading position for the style furnace under tests and operation of the furnace initiated. The furnace shall automatically process the charge through the preselected cycle sequence and record the temperature and time cycles. At the end of the one hour soak period, temperature of the heating chamber shall be within 10°F of the initial temperature. The furnace shall them be operated for at least 8 hours with maximum production loads of iron or steel. The gross weight of each load shall be equivalent to not less than 125 pounds/ square foot of heating chamber area and a different cycle sequence shall be selected for each load. Failure of the furnace to operate and comply with all requirements specified herein or any evidence of damage to the furnace or components shall be cause for rejection.

4.6.5 Temperature uniformity test. The furnace shall be loaded with a production type charge equivalent to not less than 80 percent of maximum load, and a chamber temperature survey performed to determine conformance with the uniformity requirements specified for the furnace. A minimum of nine test locations shall be used, with the thermocouple located in each corner of the chamber and one thermocouple located within six inches of the control thermocouple. The survey shall be taken with equilibrium established at minimum operating temperature specified for the furnace and under equilibrium conditions at each 100°F increase in temperature through the operating temperature range of the furnace. At least three survey readings shall be taken at intervals of five to ten minutes at each 100°F change in temperature. The maximum temperature variation between the control point and all points under test shall not exceed ± 15°F and the temperature variation between set point on the temperature controller and all points under test shall not exceed ±25°F.

* 4.6.6 Electromagnetic interference control test. Furnaces having electromagnetic interference control shall be tested in accordance with the applicable requirements of MIL-STD-461 to determine compliance with 3.4.3.

* 4.7 Inspection of packaging. Packaging shall be inspected to determine compliance to the requirements of Section 5.

5. PACKAGING

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* 5.1 Preservation, packing and marking. Preservation, packing, and marking shall be in accordance with MIL-F-3296. Level of preservation, and packing shall be as specified (see 6.2.1).

6.1 NOTES

6.1 Intended use. Furnaces covered by this specification are intended primarily for use with controlled atmosphere in such metal heat treating processes as carburizing, carbonitriding, hardening, normalizing, and annealing, with optional protective atmosphere cooling or oil quenching.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type, style, and class required (see 1.2).

c. Size required (width, length, height) (see 1.2.1 and 3.6).

d. Operating temperature range, if different (see 3.2).

e. Powered loading and unloading tables, when required (see 3.2.3).

f. Front loading purge vestibule (Type II and Style B) when required (see 3.4.2).

g. Fuel supply (BTU rating and pressure), if different (Class 1 furnace only) (see 3.2.5).

h. Electrical power supply, if different (Class 2 furnace only) (see 3.2.6).

i. Additional safety requirement, when required (see 3.2.7).

j. Motor operating voltage, if different (see 3.4.1).

k. Electromagnetic interference control, when required (see 3.4.3).

l. Air cooled heat exchanger, when required (see 3.7.7).

m. Program timers and cycles, if different (see 3.7.13).

n. Atmosphere gas metering panel, when required (see 3.8).

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o . Nameplate, if different (see 3.9).

p . Level of preservation and packing required (see 5.1).

* 6.3 Contract data requirements. Required technical data such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

* 6.4 First article. The furnace shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. A first article comprises a preproduction item or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examination, tests and approval of the first article.

* 6.5 Safety and health determination. In order that equipment integrated into the user's operational environment will comply with the standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning guarded the point(s) of operation, and other safety and health requirements should be specified.

* 6.6 Mercury prohibited. When the use of mercury of its compounds in the furnace and its components and accessories is to be prohibited, procurement documents should include a statement to that effect.

6.7 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army- AL
Navy - SH
Air Force - 99

Review activities:
Army - AL, EL
Navy - SH
Air Force - 99
DLA - GS

User activities:
Navy - MC, AS, OS
Army - SM

Preparing activity:
DLA - IP

Project Number:
3424-0097

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